PROJECT TITLE

ES3701 - Project Phase I

CyberGraph Intelligence Platform — Graph-Powered Threat Analysis Using RAG

Presenting By

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Name of the Guide

Abstract

- **Problem:** Security analysts face fragmented threat data across CVE databases, ATT&CK mappings, and IOC feeds, making multi-hop questions like "which unpatched systems are at risk from APT29?" slow and manual to answer.
- **Solution:** CyberGraph models threat actors, CVEs, malware, and techniques as a connected Neo4j knowledge graph, using Graph RAG to traverse relationships and let an LLM synthesize analyst-grade answers.
- **Impact:** Built entirely on free public data sources (NVD, MITRE ATT&CK, CISA KEV, OTX), it delivers 10 modular capabilities — from vulnerability prioritization to incident response — through a single natural-language interface.

Data collection and analysis

- **Data Collection**
- **Sources:** Ingests data from 6+ free public feeds — NVD/NIST (CVEs), MITRE ATT&CK (techniques/actors), CAPEC/CWE, AlienVault OTX (IOCs), and CISA KEV (exploited vulnerabilities).
- **Pipeline:** A Python + Celery ETL pipeline handles bulk historical loads plus incremental refresh (daily for NVD/KEV, hourly for OTX pulses).
- **Normalization:** Raw feeds are parsed, deduplicated, and normalized (CPE strings, STIX bundles, IOC dedup) before loading into Neo4j via idempotent MERGE operations.

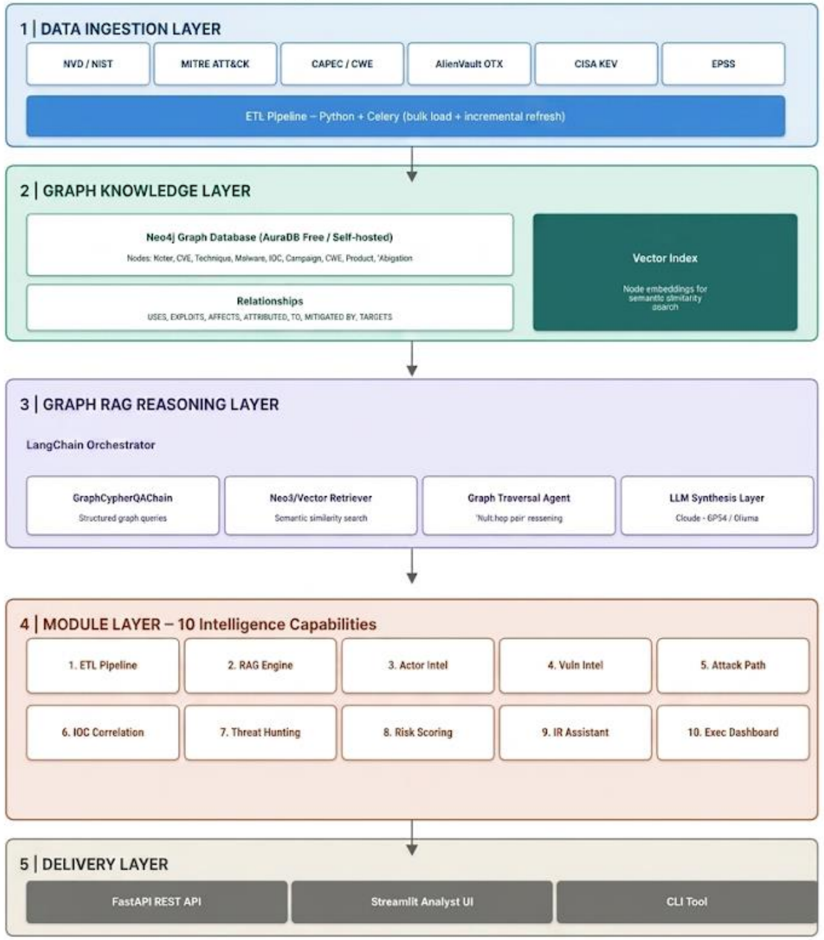
- **Analysis**
- **Graph Traversal:** LangChain's GraphCypherQAChain converts natural-language questions into Cypher queries, enabling multi-hop reasoning across actors, techniques, and CVEs.
- **Hybrid Retrieval:** Combines structured graph traversal with vector similarity search (Neo4j vector index) to handle both precise and semantic/fuzzy queries.
- **Scoring & Synthesis:** Computes composite metrics like the ThreatPriority Score (CVSS + KEV + EPSS + actor/malware exposure) and synthesizes graph evidence into LLM-generated, analyst-ready answers.

Methodology and System Design

- **Proposed Methodology**
- **Graph-centric modeling:** Represent the cybersecurity domain (threat actors, CVEs, malware, techniques) as an interconnected Neo4j knowledge graph instead of flat, siloed data tables.
- **Retrieval-augmented reasoning:** Combine structured Cypher graph traversal with vector similarity search, letting an LLM synthesize multi-hop, relationship-aware answers rather than isolated facts.
- **Modular, phased implementation:** Build ingestion, graph, and RAG layers first, then layer 10 independent intelligence modules on top — enabling iterative development and testing across a 6-week build plan.

CyberGraph Intelligence Platform

System Architecture — Detailed View



Data flows top-down: public feeds -> graph knowledge base -> RAG reasoning -> 10 modules -> analyst-facing delivery

Built with Neo4j · LangChain · Python · MITRE ATT&CK · NVD/NIST · AlienVault OTX · CISA KEV

- **Selection of Algorithms, Tools, and Technologies**
- **Neo4j 5.x (Community/AuraDB Free):** Chosen as the graph database for native multi-hop traversal, Cypher querying, GDS algorithms (shortest path, centrality), and built-in vector indexing.
- **LangChain (GraphCypherQAChain + Neo4jVector):** Orchestrates the RAG pipeline, converting natural language into Cypher queries and merging results with semantic vector search.
- **Claude 3.5 Sonnet / GPT-4o / Ollama:** Pluggable LLM backend for synthesis — Claude is recommended for long-context reasoning, with Ollama as a fully offline, air-gapped alternative.

- **sentence-transformers (all-MiniLM-L6-v2):** A free, local embedding model used to generate vector representations of node descriptions, avoiding API costs.
- **Celery + Redis:** Handles scheduled and asynchronous ETL tasks — daily NVD/CISA refreshes, hourly OTX pulse ingestion — ensuring the graph stays current.
- **FastAPI + Streamlit:** FastAPI exposes all 10 modules as REST endpoints with auto-generated docs, while Streamlit provides a rapid, no-frontend-code analyst UI with graph visualization (pyvis).

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Thank You!